

Artificial Intelligence in Software Requirements Engineering: State-of-the-Art

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Abstract—Requirements Engineering (RE) is a very important activity in the software development life cycle. Poorly executed RE steps can result in poor quality software and expensive maintenance cost. Although researchers have previously related and applied artificial intelligence (AI) to RE, little is known about the specific role of AI in RE process. In particular, there are insufficient understandings about how AI should be incorporated in the RE process to produce high quality, clear and detailed requirements. In this paper, we present the current state-of-the-art of AI in RE. We reviewed the literature published between January 2015 to December 2021 in order to understand how the state of the art of AI branches such as machine learning, classification, and natural language processing (NLP) has advanced the field of RE. Each recent study is summarized and the advancement to the RE field is presented. There is an apparent direction of applying NLP techniques and supervised learning techniques such as classification to requirements documents. This study provides a summary and direction of the AI applications in the field of RE.

Index Terms—machine learning, artificial intelligence, requirements engineering, software engineering

I. INTRODUCTION

Artificial intelligence (AI) has been applied to various fields from medicine to agriculture to manufacturing and, especially, to engineering. Particularly in computing, AI has been applied to various fields such as data mining, speech and language processing, robotics, image recognition, visual analytics and cloud computing [1]. In recent years, software requirements engineering (RE) which is a subfield of software engineering (SE) has gained attention for AI applications. The RE field has evolved in the past few decades as the need for understanding and analyzing a user's need increases to produce quality software [2]. In the early days of SE, the problem with lack of requirements analysis has become more and more apparent. The software created without identifying the needs result in poorly designed, delayed, or even canceled products [3]. As systems' complexity increases and more research is put into requirements, various techniques and processes have been identified and implemented to the art of requirements analysis.

There is a recent trend of incorporating AI into engineering processes, including RE. This paper aims to understand how AI techniques have been proposed for the steps in the RE process. Literature published between January 2015 to December 2021 was reviewed to understand how the state of the art of AI

branches such as machine learning, classification, and natural language processing has advanced the field of RE.

Section II and III in this paper provides the background information necessary to understand the terminologies from the literature. The review and selection methodology used in this research is presented in section IV. Section V presents the result of this study and identifies the AI techniques applied to each RE step. Section VI presents the conclusion and recommendation for future work.

II. REQUIREMENTS ENGINEERING

Requirements engineering (RE) refers to a set of activities concerned with identifying and communicating the purpose of a software-intensive system and the contexts in which it will be used [4] [5]. A consolidated definition from IEEE for a software requirement is a documented condition or capability that a system or system component meets or possesses to solve a contractual, standard, or specification problem or objective [6]. RE processes also establish the relationships between requirements and the software system and track the evolution over time. The goal of this process is to document the understanding or discovery of what the user really needs.

The RE process begins with requirement elicitation and analysis. Requirements from stakeholders are collected through methods such as interviews, observation, domain analysis, prototyping, etc [7]. The requirements are not simply collected but also analyzed. Requirements from stakeholders often come in natural language form. Further analysis is needed to discover and reveal the problem or ambiguities that need to be solved. Any insufficiency in the analysis will negatively impact all the steps after including development and maintenance [8]. In requirements specification activity, the analyzed requirements are formally documented and modeled [9]. The requirements are represented as entity models and relationships are visualized in diagrams such as entity-relationship diagrams, data flow diagrams, sequence diagrams, etc. These are all recorded into the software requirements specification (SRS) document.

Requirement validation confirms that the software specification meets the user requirements and ensures that the implementation can be traced to the requirements. Validating

requirements results in significant cost reduction and increase in software quality [10].

III. ARTIFICIAL INTELLIGENCE

In computer science, AI is defined as a system that acts humanly, thinks humanly, acts rationally, and thinks rationally [11]. AI has been applied to various scientific fields to propel advancements in new research and discoveries. Due to advancements in processor speed, AI systems can observe, perceive and process exponentially faster and significantly better than humans. This can lead to unexpected discoveries, designs, or performance improvements since AI is not bound by the human thinking process. The field of AI has evolved to multiple subsets such as robotics, machine vision, machine learning, deep learning, natural language processing, etc. In this research, we primarily focus on two AI techniques: machine learning and natural language processing.

A. Machine Learning

Machine Learning (ML) is a field in AI that allows a computer system to learn from a dataset without knowing the dataset in advance [12]. This technique utilizes data mining to learn and recognize patterns in large amounts of data. Within the field of ML, there are two main types of tasks such as supervised learning and unsupervised learning. In unsupervised learning, the ML algorithm makes sense of the data by only being provided the input data. Clustering and Neural network is an example of unsupervised learning. Whereas, supervised learning, the ML algorithm is provided a set of known input and labeled output. The algorithm is first "trained" to make predictions. Classification and regression is an example of supervised learning.

B. Natural Language Processing

Natural Language Processing (NLP) is a computational technique that analyzes linguistic text by an AI system to process natural language text or speech [13]. NLP applies various levels of linguistic analysis to the target such as tokenization, part-of-speech tagging, decomposition, spelling/grammar error recovery, etc. In terms of processing input, the goal of this AI technique is to draw inferences or derive useful information from the natural language text.

IV. METHODOLOGY

This study aims to review recent advancements in AI for RE. In order to conduct our review, we identified a set of search strings and logical operators connecting those strings. After performing trial searches for several rounds, the strings used in our study are

- S1: requirements engineering
- S2: artificial intelligence
- S3: machine learning

The connectors used are logical-and (AND) and logical-or (OR). The string formed therefore is (S1 AND S2) OR (S1 AND S3). The search string has two parts: one related to AI and other related to ML. To ensure that the literature

is from recent years, the publication period criteria were restricted from 2015 to 2021. The sources of our search process include IEEE Xplore, ACM Digital Library and ScienceDirect databases, where a majority of computer science and software engineering papers are published. Our search is only limited to peer-reviewed journals and conference papers. This means that in our inclusion criteria, we ignored news releases, technical reports and other non peer-reviewed articles. The top 100 results for each search were kept for screening. This resulted in a total of 600 papers. From the 600 papers, 25 papers were found to be duplicated. The literature eligibility strategy involves filtering papers that contain the phrases "artificial intelligence" or "machine learning" in the keywords section and ensuring that requirement engineering is discussed in the title or the abstract. This filters the number of papers to 171 candidate papers. Finally, the abstract and content of each paper were evaluated manually for quality. This confirms that the papers used in this research is relevant to the scope of this study. Using this search strategy, 21 papers were found to be relevant to be used as a representation for the state-of-art of AI in the RE process. This process is illustrated in figure 1. The papers analyzed in this study are shown in Table I. Figure 2 shows the trends in AI for RE from 2015 to 2021.

V. ARTIFICIAL INTELLIGENCE IN THE REQUIREMENTS ENGINEERING

A. Requirements Elicitation and Analysis

Requirements elicitation and analysis is an important activity in the RE process. In this activity, requirements engineers work with clients and end-users to find more about the application domain, what functionalities the system should provide, constraints, and the required performance and efficiency of the system. Requirements are often collected from the stakeholders by requirements engineers through interviews. This manual process, however, is dependent on the experience and domain knowledge of the engineer. Therefore, it relies on the requirements engineer performing the interview to ask the right questions based on their knowledge.

Rajender et al. [22] proposed a method using an AI chatbot with NLP to interact with the users to capture requirements. The knowledge base is integrated with the chatbot to provide domain requirements from various industries such as banking, healthcare, entertainment, etc. This ensures that the chatbot will ask the necessary questions to formulate an adequate, complete, and clear requirements. Since interacting with human users, the chatbot employs entity extraction, intent classification and NLP techniques to understand the entities and intents in the user's responses. The elicited requirements are classified into functional and non-functional categories using Support Vector Machines (SVM) and Multinomial Naïve Bayes (MNB) and stored accordingly. These two techniques achieved 88% and 91% accuracy respectively.

Convolutional Neural Network (CNN) was applied by Winkler et al. [23] to classify specific content as a requirement or non-requirement. The CNN consists of a filter layer and an association layer. The filter layer filters out the input

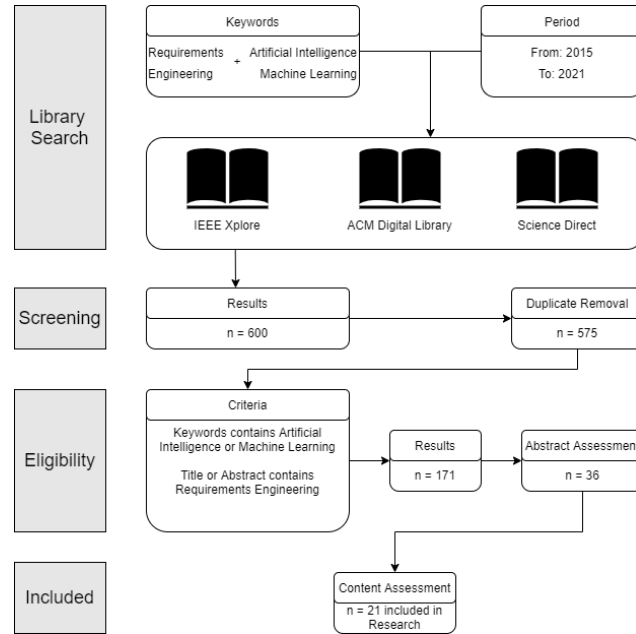


Fig. 1. Literature Search and Selection Process

Authors	Published On	Year	AI Task	RE Step
Nguyen et al. [14]	ACM	2015	Classifier, NLP	Requirements Validation
Abdouli et al. [15]	IEEE	2016	NLP	Requirements Specification
Abad et al. [16]	IEEE	2017	Classifier, NLP	Requirements Specification
Lu and Liang [17]	ACM	2017	Classifier, NLP	Requirements Elicitation
Osman and Zaharin [18]	IEEE	2018	Classifier, NLP	Requirements Validation
Alrumaih et al. [19]	IEEE	2018	Classifier, NLP	Requirements Specification
Qayyum and Qureshi [20]	ACM	2018	Classifier	Requirements Specification
Haider et al. [21]	IEEE	2019	Classifier	Requirements Specification
Rajender et al. [22]	IEEE	2019	Classifier, NLP	Requirements Elicitation
Winkler et al. [23]	IEEE	2019	Neural Network	Requirements Elicitation
Dalpiaz et al. [24]	IEEE	2019	Classifier, NLP	Requirements Specification
Ebrahimi and Barforoush [25]	IEEE	2019	Classification, NLP	Requirements Elicitation
Khan et al. [26]	IEEE	2019	Classifier, NLP	Requirements Elicitation
Lyutov et al. [27]	ScienceDirect	2019	Classifier	Requirements Specification
Binkhonain and Zhao	ScienceDirect	2019	Classifier	Requirements Specification
Tiwari et al. [28]	IEEE	2020	NLP	Requirements Specification
Hey et al. [29]	IEEE	2020	Classifier, NLP	Requirements Specification
Panichella and Ruiz [30]	IEEE	2020	Classifier	Requirements Elicitation
Singh et al. [31]	IEEE	2021	Classifier, NLP	Requirements Validation
Akay and Kim [32]	ScienceDirect	2021	NLP	Requirements Elicitation

TABLE I
LITERATURE ON AI APPLICATIONS IN THE RE PROCESS

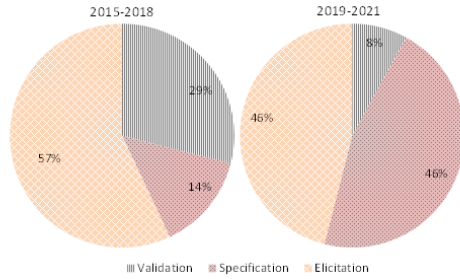


Fig. 2. Applications of AI in different RE activities

by patterns and the association layer associates the input to a label. The classifier is trained with 35,000 pre-labeled data. The researchers found that after optimization, CNN can achieve up to 98% recall but a reduction in the precision value. It was concluded that there is a need to find a right balance between recall and precision, but this tool can assist requirement engineers to find defects in requirements that need further clarification.

Ebrahimi and Barforoush [25] took advantage of Twitter to serve as software reviews to extract useful information. NLP tasks such as replacements of numerics, URLs, hashtags, punctuation removal, stop words removal, lowercasing and stemming were performed on Twitter input data. Three different classifiers, Multinomial Naïve Bayes (MNB), Decision Tree (DT), and Random Forest (RF), were applied to the processed data. The RF classifier was found to increase in performance when more NLP processes are applied to the data. This study established that textual preprocessing tasks are important to extract useful information from tweets to contribute to the requirements elicitation process.

Crowd-user feedback such as app reviews, user forums, tweets, and issue tracking systems are used in the RE process to identify new requirements or issues. This is known as crowd requirements engineering (CrowdRE) [33] [34]. Khan et al. [26] proposed an approach to analyzing user comments on user forums to obtain requirement-related arguments. User comments are analyzed using NLP to determine whether a user supports, attack, suggest, or questions a requirement. The researchers applied Abstract Argumentation [35], Bipolar Argumentation Framework [36] and Crowd Requirements Argumentation Framework to identify opinion conflicts, inconsistency, and vagueness in the user comments. This approach classifies the comments as issues, claims, and features using MNB, SVM, RF, and Logistic Regression (LR). The SVM-TFIDF-n-gram algorithm resulted in the best performance with 83% F-score. This research established a proof-of-concept to use AI to extract useful requirement information from user forums.

Lu and Liang [17] performed a similar study on the app reviews from Apple App Store and Google Play Store. Similar NLP techniques were applied to the user reviews. The weight of each was calculated using BoW, Chi-squared (χ^2), and TF-

IDF techniques. Classifier algorithms such as Naïve Bayes, J48 and Bagging, were used to classify text into non-functional requirements. Each non-functional requirement is then classified to various types such as usability, reliability, portability and performance. AUR-BOW with Bagging achieved the highest F-measure of 71.8%, with a precision of 71.4% and a recall of 72.3% respectively, in classifying non-functional requirements from app user reviews.

Akay and Kim [32] developed a novel system called *Design Reading* that extracts a hierarchical map of functional requirements and design patterns from customer requirements. This technique uses the neural network-based NLP model for context extraction called BERT, to extract the functional requirements and design parameter recursively until all the relevant texts have been exhausted. At each recursive step, each of these questions is answered: “what is the aim?”, “how should it work?” and “what is needed?”. The Design Reading system is built to be a module of the hybrid intelligence system proposed by the authors.

Panichella and Ruiz [30] implemented a tool called Requirements-Collector that automates requirements specification and user feedback analysis. The tool aims to perform real-time audio transcription and incorporate user reviews to dynamically generate user stories and classify user reviews during the requirements elicitation session. NLP is used to process the audio transcribe and supervised ML techniques to identify functional and non-functional requirements. The goal of this tool is to assist requirement engineers during the live requirements elicitation session using AI to collect high-quality requirements.

B. Requirements Specification

The Software Requirement Specification (SRS) document records all the requirements for a software system. SRS includes both the user requirements and system requirements for a system. Software system requirements are classified as functional requirements (FR) and non-functional requirements (NFR). Various studies have been done on the automation of the classification process using AI. Abad et al. [16] investigated how automated classification of requirements into FR and NFR can be improved and how well the AI and ML approaches work in this context. The results showed that using NLP techniques to extract grammatical, temporal, and sentimental characteristics improved the performance of both classifying FR/NFR and sub-classifying NFR into other categories. However, Binkhonanin and Zhao [37] found that, for classifying NFRs, NLP’s stemming and lemmatization reduce the performance of ML algorithms. Both studies showed that Naïve Bayes (NB) classifier can achieve high accuracy for NFRs.

Dalpia et al. [24] proposed that classifiers for FR and NFR can perform well with only high-level linguistic features and dependency parsing. The researchers produced a new manually annotated dataset based on the requirements in the PROMISE dataset. They found that interpretable ML algorithms, such as rule-based classification, can be just as effective on a dataset

that has high-level linguistic dependencies when compared to ML on a dataset with low-level linguistics such as n-grams. Hey et al. [29] developed an approach called NoRBERT based on BERT's transfer learning. The model is trained using relabeled PROMISE dataset and the results showed that FR and NFR classification using the high-level linguistic feature dataset can achieve high performance.

Alrumaih et al. [19] study suggested a hybrid approach of using multiple AI algorithms to achieve high performance in classifying requirements. This research provided a direction for the requirements classification tasks using AI. A multi-voting approach based on a mix of classifier algorithms showed a high level of accuracy for assigning requirements into right departments [27].

Tiwari et al. [28] proposed using NLP's NER technique to extract named entities to generate use case elements. NLP techniques can also be used to generate UML diagrams such as ER, object, class diagrams from the SRS document. Abdouli et al. [15] found high performance in implementations using AI to transform the textual requirements to an ontology for diagram generation. The requirements specification phase also involves requirements prioritization to determine which requirements are more important to be implemented sooner. One approach proposed by Haider et al [21], is using the J48 classifier to rank the new requirement based on previous similar requirements. This technique also achieves traceability as it identifies other similar requirements.

C. Requirements Validation

Requirements validation ensures the requirements are not ambiguous and the SRS document does not have any issues that may cause poor quality in the subsequent software coding phase. Osman and Zaharin [18] used NLP to extract feature words that influence a requirement to be ambiguous. The requirements are then classified as ambiguous or unambiguous. They found that Random Forest (RF) classifier produced the best results for classifying ambiguous or unambiguous requirements. Singh et al. [31] conducted a similar study but found that the J48 DT produced high accuracy classifying requirements into ambiguous and unambiguous classes from the SRS document. Nguyen et al. [14] proposed a novel approach using NLP's linguistic analysis, coreference resolution, and syntactic and dependency parsing, to extract artifacts and relationships from the SRS document. The extracted information is then used to produce goal-use case models to validate the correctness of the SRS documentation.

VI. CONCLUSION

In this survey paper, we showed that there has been an extensive amount of research on incorporating AI techniques in the RE process. Given the current state of research of AI in RE, researchers should consider opportunities to extend and improve present research in an integrative fashion. The current state-of-the-art shows that applying suitable ML and NLP techniques to textual requirements is important to derive further meaning such as functional vs non-functional

requirements, ambiguous vs unambiguous requirements, type of non-functional requirements, etc. As pointed out in some of the research studies reviewed in this paper, some NLP techniques improve the performance of ML while other NLP techniques degrade the performance. It is evident that there is a high potential for the applications of AI in RE. AI can help requirement engineers to focus on the creativity in the RE process rather than the mundane tasks of quality checking, ambiguity, classifying, etc. With the advancement in AI, the paradigm in RE should be expected to shift to integrate AI into the RE process.

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